Carola Bonamico

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EDUCATION

Polytechnic University of Turin — Turin, IT
MSc in Computer Engineering – Software — *Sep 2024 – Oct 2026*

- Final grade: 110/110
- Thesis: "From Biosignals to Words: Exploiting Novel Deep Learning Architectures for Speech Understanding"
- Main courses: Data Science and Database Technology, Software Engineering I/II, Web Applications I, Large Language Models for Software Engineering, Human Computer Interaction, Computer Architectures, System and Device Programming

University of Pavia — Pavia, IT
BSc in Biomedical Engineering — *Sep 2020 – Apr 2024*

- Thesis: "Literature review on the impact of sleep quality on glycemic patterns in patients with type 1 diabetes"
- Main courses: Biomedical Data Processing, Models of Biological Systems

I.I.S. Cellini – Liceo Linguistico Alberti — Valenza, IT
Secondary School Diploma in Modern Languages — *Sep 2015 – Jun 2020*

- Languages: English, French, German

EXPERIENCE

Huawei Seeds for the Future Italy 2026 — Beijing, Shanghai, CN
Program Participant — *Sep 2026 – Sep 2026*

- Selected as one of **6 students out of over 100 applicants** for Huawei's fully-funded flagship talent program in Beijing and Shanghai.
- Engaged in a full-immersion training on technology, innovation, and sustainability, designed to combine academic merit and entrepreneurial spirit within the advanced Chinese ecosystem.

ETH Zurich — Zurich, CH
Master's Thesis Student — *Mar 2026 – Aug 2026*

- Grade: 5.5/6.0
- Conducted Master's thesis project at the Integrated Systems Laboratory (IIS) under supervision of Prof. Dr. Luca Benini, with focus on developing edge-constrained **Deep Learning** models for silent speech recognition from electromyographic signals.
- Reached **85.8%** and **82.1%** accuracy (vocalised, silent) in classification and **16.3%** and **22.4%** Word Error Rate (WER) (vocalised, silent) in continuous recognition.
- Designed and collected a multimodal, multi-session dataset from **7 subjects** for **40 hours** of synchronised EMG, EEG and audio.

Squadra Corse PoliTo — Turin, IT
Software Developer — *Aug 2025 – Nov 2025*

- Built a TUI in Go to send and inspect CAN messages over the SocketCAN interface for **debugging and troubleshooting**.

TECHNICAL SKILLS

Programming languages: C, Rust, Go, JavaScript, TypeScript, Python, ARM Assembly, MATLAB
Deep Learning: PyTorch, NumPy, Pandas, Scikit-learn
Libraries and frameworks: Node.js, React, React Native, TypeORM, Diesel
Databases: Oracle SQL, MongoDB, MySQL, SQLite, PostgreSQL
Tools and software: Git, GitHub, GitLab, Docker, Jupyter Notebook, LabVIEW
Other: REST APIs, Agile, UML, HTML, CSS

RELEVANT PROJECTS

SilentWear
Open-source project based on DL models for silent speech recognition from EMG signals. Extended the existing CNN with a BiLSTM and a **Transformer**, added CTC decoding for direct EMG-to-text recognition alongside closed-set classification, and implemented an EMG-based automatic speech onset detection algorithm.

Autostar Motorsport Web App
Responsive web application for a motorsport company, built with **Node.js** and **React**. Implements a client-side architecture in **JavaScript**, **GDPR** compliance through Iubenda and modular React components styled with **CSS** and **AOS**.

Participium
Full-stack platform for civic issue reporting in **TypeScript/JavaScript** (**Node.js** and web frontend), with role-based flows, geolocation and photo upload. Includes **Telegram** notifications and **SonarQube** for static analysis and continuous code quality monitoring.

Rust Chat
Real-time group messaging application built with **Rust** and **React**. Supports private chats, groups, JWT authentication, WebSocket communication and **Docker** deployment.

Geocontrol
Dockerised software monitoring system for managing geological, meteorological and environmental sensors. Built in **TypeScript** using the repository pattern, **TypeORM** and **Node.js**.

CERTIFICATIONS

Goethe-Zertifikat B1 (German) – Goethe-Institut Italia — *Issued: Jul 2019*
B1 Preliminary (English) – Cambridge English — *Issued: May 2018*
Delf A2 (French) – Ministère de l'Éducation nationale — *Issued: Sep 2015*